

Multi-objective Genetic Algorithm Optimization of Offshore Wind Farm Layout, Substation Placement, and Cable Grouping

Anonymous Author(s)

Abstract

Offshore wind farm (OWF) design involves tightly coupled aerodynamic and electrical decisions that are often optimized sequentially, potentially leading to sub-optimal trade-offs. This paper presents a comparative study of three evolutionary optimization paradigms for OWF co-design: (i) a single-phase NSGA-II optimizing energy production and cabling cost from the outset, (ii) a sequential approach optimizing turbine layout prior to electrical infrastructure design, and (iii) a hierarchical two-phase strategy that combines layout-focused exploration with subsequent multi-objective refinement. All methods are evaluated on a commonly adopted OWF toy problem with increasing farm sizes. Results show that the proposed two-phase strategy consistently achieves higher-quality Pareto fronts, balancing net annual energy production and cabling cost more effectively than purely sequential or single-phase approaches, while maintaining reasonable computational overhead. The study highlights the impact of problem decomposition and optimization strategy on solution quality in real-world evolutionary design problems.

CCS Concepts

• **Computing methodologies** → Genetic algorithms; • **Hardware** → Resource optimization.

Keywords

Offshore Wind Farms, Multi-objective Optimization, Cable Routing, NSGA-II, Substation Placement

1 Introduction

Offshore wind power is a key component of the global energy transition, benefiting from high and stable wind resources and increasing deployment worldwide. Despite its technological maturity, offshore wind farms remain capital intensive, with subsea electrical infrastructure accounting for approximately 11% to 15% of total project costs [8, 23]. A central challenge arises from the strong coupling between turbine layout and electrical collection system design: turbine repositioning to mitigate wake losses directly affects cable length, Joule losses, and routing feasibility.

Most Wind Farm Layout Optimization (WFLO) studies primarily focus on maximizing Annual Energy Production (AEP) through wake mitigation [1, 4]. While this approach yields aerodynamically efficient layouts, it may lead to economically unfavorable solutions once electrical infrastructure is considered. Conversely, recent advances in the Wind Farm Cable Routing Problem (WFCRP) typically assume fixed turbine and substation locations, addressing electrical optimization as a secondary and decoupled task [16, 23]. As a

result, the interaction between aerodynamic layout decisions and electrical design choices remains insufficiently explored.

Integrated optimization frameworks have emerged to address this limitation, demonstrating that simultaneous consideration of layout and electrical infrastructure can reduce overall project costs [12, 25]. However, two gaps persist in the literature. First, most approaches treat the number of collector strings as fixed, limiting topological flexibility. Second, the transition between rapid layout exploration and detailed electrical refinement is often handled in a disjointed manner, increasing sensitivity to local optima and computational burden.

This work extends the modular genetic algorithm framework proposed in [4, 22] toward an integrated multi-objective co-design setting. The offshore wind farm design problem is formulated with $2n + 3$ decision variables, where n denotes the number of turbines, capturing turbine positions, substation location, and cable topology. The framework explicitly explores the trade-off between maximizing Net AEP—accounting for wake and electrical losses—and minimizing total cabling cost.

Rather than proposing a new evolutionary algorithm, this paper presents a comparative analysis of three evolutionary optimization paradigms: (i) a single-phase NSGA-II optimizing layout and electrical variables simultaneously [5], (ii) a sequential two-phase strategy in which layout optimization precedes electrical design, and (iii) a hierarchical two-phase approach that combines layout-focused exploration with subsequent multi-objective refinement. The main contributions of this work are summarized as follows:

- A two-phase hierarchical optimization strategy that leverages mono-objective layout exploration to guide subsequent multi-objective co-optimization.
- A phase-bridging population initialization mechanism that seeds high-quality layouts while diversifying substation locations to preserve exploration.
- A deterministic angular sector grouping strategy that enforces planar, non-overlapping cable strings without the need for geometric repair heuristics.

2 Related Work

The optimization of offshore wind farms (OWFs) has evolved from simple geometric layouts to high-dimensional, multidisciplinary design problems involving aerodynamic, electrical, and economic objectives. This section reviews prior work on wind farm layout optimization, cable routing, and integrated co-design approaches, with emphasis on evolutionary and meta-heuristic methods.

2.1 Wind Farm Layout Optimization (WFLO)

Wind Farm Layout Optimization (WFLO) aims to mitigate wake-induced energy losses, which significantly degrade the performance of downstream turbines. Classical analytical wake models, such as the Jensen model [9, 14] and the Gaussian formulations proposed

by Bastankhah and Porté-Agel [2, 3], are widely adopted due to their computational efficiency and suitability for optimization loops. These models are commonly embedded in Genetic Algorithms and hybrid meta-heuristics to address the non-convex turbine placement problem [7, 20]. Benchmark-driven studies, such as those based on the IEA Task 37 framework [1], have further standardized comparative evaluations of WFLO methods. However, most WFLO approaches focus exclusively on maximizing gross AEP, neglecting electrical infrastructure and leading to layouts that may be energetically optimal but economically inefficient [4].

2.2 Wind Farm Cable Routing Problem (WFCRP)

The Wind Farm Cable Routing Problem (WFCRP) is an NP-hard combinatorial problem that seeks to minimize the cost of connecting turbines to offshore substations while respecting electrical and geometric constraints. Early solutions relied on Mixed-Integer Linear Programming (MILP) formulations [6] or heuristic algorithms such as the Esau-Williams method [11]. A central difficulty in WFCRP is the enforcement of non-crossing cable layouts, which has motivated the use of geometric repair heuristics, detours, or delayed embedding strategies [23, 26]. Recent work has explored multi-agent and swarm-based meta-heuristics to improve scalability [16, 27], as well as topology-driven designs such as ring or hybrid networks [21]. Nevertheless, most WFCRP studies assume fixed turbine and substation locations, treating routing as a secondary optimization step.

2.3 Integrated Co-Design and Substation Placement

Integrated optimization of turbine layout and electrical infrastructure has gained increasing attention as a means to reduce overall project costs. Several studies show that optimizing substation location jointly with cable routing yields measurable cost reductions compared to sequential approaches [12, 13, 17]. Other works incorporate cable sizing, electrical losses, and cost models directly into the optimization loop [18, 25]. Despite these advances, two limitations remain prevalent in the literature. First, the number of collector strings is typically predefined rather than treated as a decision variable. Second, few studies systematically compare different evolutionary optimization strategies and problem decompositions for the fully coupled layout-infrastructure problem, particularly in terms of convergence behavior and Pareto front diversity.

3 Methodology

This work formulates offshore wind farm (OWF) design as a multi-objective co-design problem coupling aerodynamic performance and electrical infrastructure cost [12, 25]. Three evolutionary optimization strategies are compared under a unified modeling framework: a single-phase NSGA-II baseline [5], a sequential optimization approach [13, 17], and a two-phase hierarchical framework, originally proposed in [4, 22], extended in this work to include electrical cabling.

3.1 Problem Definition and Decision Variables

The OWF co-design problem is represented by the decision vector

$$\mathbf{X} = [x_1, y_1, \dots, x_n, y_n, G_{norm}, SS_x, SS_y], \quad (1)$$

where (x_i, y_i) denote turbine coordinates, (SS_x, SS_y) is the offshore substation position, and $G_{norm} \in [0, 1]$ is a continuous grouping gene mapped to an integer number of collector strings $N_{groups} \in [N_{min}, N_{max}]$. This encoding enables standard continuous evolutionary operators while implicitly handling the mixed-integer nature of the design problem [7].

3.2 Optimization Strategies

Three optimization paradigms are evaluated:

Baseline (Single-Phase NSGA-II): A pure NSGA-II formulation [5] jointly optimizes Net AEP and cabling CAPEX from the initial population, evolving all decision variables simultaneously.

Sequential Optimization: A single-objective GA first optimizes turbine placement to maximize gross AEP. The resulting best layout is then fixed, and a second GA optimizes substation location and cable grouping to minimize electrical cost.

Proposed Two-Phase Hierarchical Strategy: This approach, illustrated in Fig. 1, handles the high dimensionality of the co-design problem through a structured transition between energetic and infrastructure optimization. Phase 1 applies a single-objective GA to maximize gross AEP over the reduced $2n$ -dimensional layout space, ignoring electrical effects to promote rapid exploration. As depicted in the transition bridge of the framework, Phase 2 initializes its population using the best layouts from Phase 1 (Smart Seeding). Substation positions are initialized at each layout's centroid and randomly perturbed to preserve diversity. NSGA-II then optimizes Net AEP and CAPEX over the full $2n + 3$ decision vector.

3.3 Deterministic Cabling via Strict Angular Partitioning

To construct feasible, non-crossing electrical collection systems without geometric repair [23, 26], a deterministic topology synthesis mechanism termed *Strict Angular Partitioning (SAP)* is adopted. This mechanism, integrated into the multi-objective evaluation loop shown in Fig. 1, transforms the expanded decision vector into a valid collection network. For a given substation position, turbines are sorted by polar angle, partitioned into contiguous angular sectors according to N_{groups} , and connected radially from the outermost turbines toward the substation. This guarantees planar cable layouts and enables efficient evaluation within the evolutionary loop.

3.4 Objective Functions

The three optimization strategies employ different objective function configurations across their respective phases. The fundamental objectives considered are:

Objective 1 – Maximize Net AEP:

$$f_1(\mathbf{X}) = \text{AEP}_{gross} - \mathcal{L}_{joule}, \quad (2)$$

where gross AEP is computed using a Gaussian wake model [2, 3] and $\mathcal{L}_{joule}$ represents electrical Joule losses in the collection system.

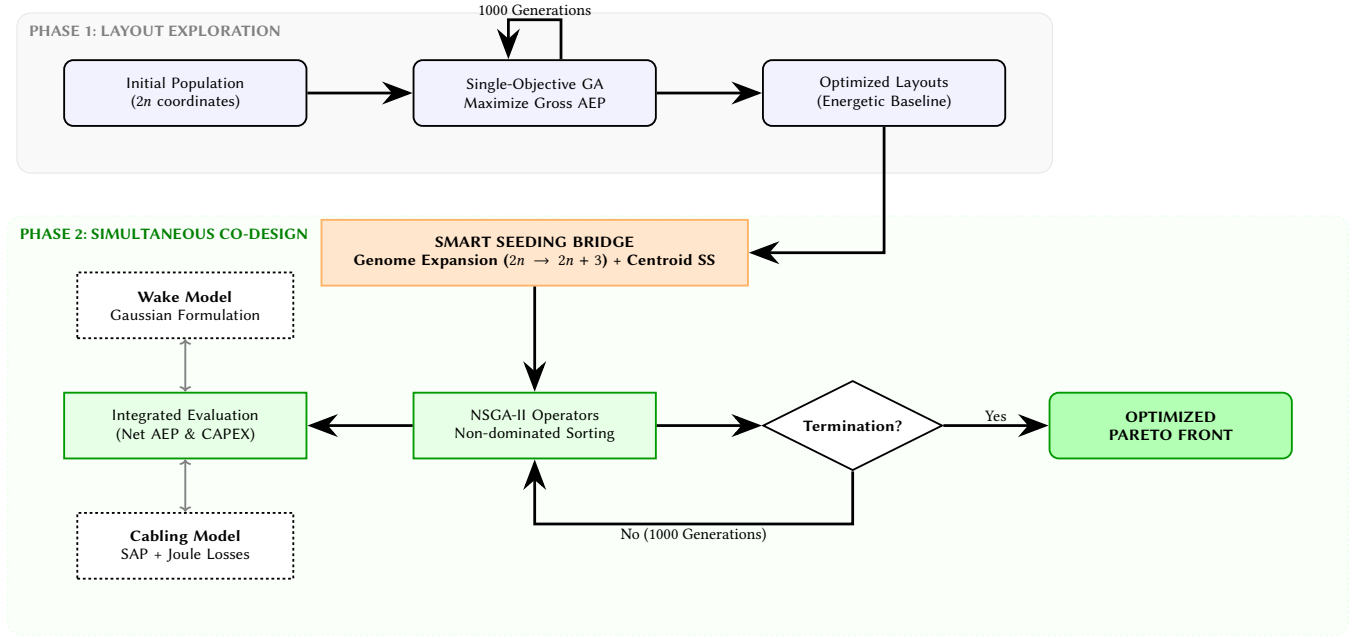


Figure 1: Proposed hierarchical evolutionary framework. Phase 1 performs an energetic search in the $2n$ spatial domain, while Phase 2 executes the multidisciplinary co-design loop using the Smart Seeding mechanism to bridge both optimization stages.

This objective requires simultaneous evaluation of turbine layout and electrical infrastructure, as cable routing determines $\mathcal{L}_{joule}$.

Objective 2 – Minimize Cabling CAPEX:

$$f_2(\mathbf{X}) = \sum_{c=1}^{N_{cables}} L_c \cdot C(CSA_{uniform}), \quad (3)$$

where L_c denotes cable lengths and $C(CSA_{uniform})$ is the cost of a uniformly selected commercial cable cross-section [18].

Baseline (Single-Phase NSGA-II): Both objectives f_1 (Net AEP) and f_2 (Cabling CAPEX) are optimized simultaneously from the initial population using NSGA-II [5], evolving all decision variables ($2n + 3$) concurrently throughout the entire optimization process. Since cable routing is evaluated at each generation, $\mathcal{L}_{joule}$ is explicitly accounted for in f_1 .

Sequential Optimization: Phase 1 applies a single-objective GA to maximize gross AEP over the $2n$ -dimensional layout space, ignoring electrical infrastructure entirely. The objective function is simply AEP_{gross} (without $\mathcal{L}_{joule}$), as cable routing is not evaluated at this stage. The resulting best layout is then fixed, and Phase 2 employs a second single-objective GA to minimize f_2 (cabling CAPEX) by optimizing substation location and cable grouping variables only, with turbine positions held constant.

Proposed Two-Phase Hierarchical Strategy: Phase 1 applies a single-objective GA to maximize gross AEP over the reduced $2n$ -dimensional layout space, using the same objective as Sequential Phase 1 (gross AEP only, without electrical losses). Phase 2 initializes its population using the best layouts from Phase 1 (Smart Seeding) and employs NSGA-II [5] to simultaneously optimize both f_1 (Net AEP, including $\mathcal{L}_{joule}$) and f_2 (Cabling CAPEX) over the

full $2n + 3$ decision vector, enabling explicit exploration of energy–cost trade-offs while allowing turbine repositioning to account for electrical infrastructure effects.

3.5 Case Study Configuration

All strategies are evaluated on a reference offshore wind farm design problem derived from the IEA Task 37 benchmark [1]. A fixed wind regime with constant wind speed of 9.8 m/s (rated speed) and a standard 3.35 MW reference turbine are adopted. Experiments are conducted for three problem scales (16, 36, and 64 turbines) within a fixed circular site of 5000 m radius, with a minimum inter-turbine distance constraint of 2 rotor diameters (260 m). This configuration enables controlled comparison of solution quality, convergence behavior, and computational cost across increasing problem dimensionality.

3.6 Experimental Design and Evaluation Protocol

The goal of the experimental design is not only to assess solution quality, but to analyze how different evolutionary optimization paradigms affect convergence behavior, diversity preservation, and trade-off exploration in a coupled aerodynamic–electrical design problem.

All optimization strategies are evaluated under identical computational budgets, population sizes, and initial conditions to ensure a fair comparison. Each strategy uses a population size of 300 individuals and equivalent computational budgets: Baseline executes 2000 generations, while Sequential and Proposed each execute 1000 generations in Phase 1 followed by 1000 generations in Phase 2, totaling 2000 generations per strategy. To ensure fair comparison, all

methods share the same global initial population for each independent run. Twenty independent runs are performed for each strategy, with each run using a unique random seed that is shared across all methods, following best practices in evolutionary computation [5, 7].

Performance is assessed using both objective-space and algorithmic metrics. In addition to Net AEP and cabling CAPEX, the following indicators are considered: (i) the number of non-dominated solutions, (ii) Pareto front spread, and (iii) hypervolume (HV). Hypervolume is adopted as the primary performance indicator due to its ability to simultaneously capture convergence and diversity, making it particularly suitable for engineering design trade-offs [5].

Computational efficiency is also reported, as scalability remains a key concern in real-world offshore wind farm design, where the dimensionality of the search space grows linearly with the number of turbines. This is especially relevant when comparing single-phase and decomposed optimization strategies operating in spaces of dimension $2n$ versus $2n + 3$.

4 Results and Discussion

This section presents a comprehensive comparison of three evolutionary optimization strategies, Baseline NSGA-II, Sequential optimization, and the Proposed two-phase hierarchical framework, evaluated across increasing problem scales (16, 36, and 64 turbines). The analysis combines quantitative multi-objective performance indicators with qualitative inspection of Pareto-optimal layouts to assess how problem decomposition, search dimensionality, and evolutionary strategy influence scalability, convergence behavior, robustness, and trade-off resolution. Similar comparative and scale-aware analyses have been identified as essential for assessing practical relevance in recent WFLO and integrated OWF design studies [10, 24].

4.1 Quantitative Performance Across Problem Scales

Table 1 summarizes the main performance indicators obtained for the three turbine densities. Across all scenarios, the Sequential strategy consistently converges to a single solution. This outcome is a direct consequence of its single-objective decomposition and fixed-layout assumption, which precludes the emergence of Pareto fronts and prevents explicit exploration of energy–cost trade-offs. Similar limitations of sequential or partially fixed approaches have been reported in earlier WFLO–WFCRP studies [6, 17].

Both the Baseline NSGA-II and the Proposed framework generate non-dominated solution sets; however, important differences emerge. For the 16-turbine scenario, the Baseline strategy achieves Net AEP ranging from 310.61 to 440.03 GWh, while the Proposed framework achieves a higher and narrower range from 429.93 to 463.06 GWh, indicating superior exploration of high-AEP regions. The Proposed framework consistently achieves higher HV values (8.99 vs 8.52×10^{12}) and greater spread (0.034 vs 0.024), reflecting improved coverage of infrastructure-driven trade-offs and greater diversity in the Pareto front.

As the problem size increases to 36 and 64 turbines, [AWAITING DATA:"scalability analysis","36 and 64 turbines"]. The Baseline

NSGA-II is expected to exhibit degradation in Pareto front quality, consistent with well-known scalability limitations of fully coupled evolutionary searches in high-dimensional design spaces [7, 19]. In contrast, the Proposed two-phase strategy is expected to maintain stable Pareto diversity and superior objective-space coverage, demonstrating increased robustness to dimensionality growth.

Table 1: Quantitative comparison of optimization strategies across turbine counts (mean $\pm$ std over 20 independent runs).

Turb.	Method	Net AEP [GWh]	CAPEX [kUSD]	Pareto Size	HV [$\times 10^{12}$]
16	Baseline	404.34 ± 8.44	190 ± 22	293 ± 6	8.52 ± 0.12
	Sequential	462.99 ± 0.43	731 ± 108	1 ± 0	8.92 ± 0.05
	Proposed	457.89 ± 2.39	657 ± 91	297 ± 4	8.99 ± 0.03
36	Baseline	TBD	TBD	TBD	TBD
	Sequential	TBD	TBD	1	TBD
	Proposed	TBD	TBD	TBD	TBD
64	Baseline	TBD	TBD	TBD	TBD
	Sequential	TBD	TBD	1	TBD
	Proposed	TBD	TBD	TBD	TBD

4.2 Pareto Front Structure and Trade-off Analysis

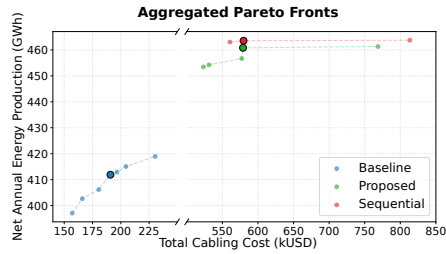
Figure 2 compares the Pareto fronts obtained by the three strategies across increasing problem scales. The Sequential strategy converges to a single solution per run, confirming its inability to expose trade-offs within each optimization run. In the aggregated plot, multiple Sequential points appear (one per independent run), but these represent different converged solutions rather than a Pareto front from a single run. The Baseline and Proposed strategies both generate Pareto fronts within each run; however, important differences emerge in their objective-space coverage. For the 16-turbine scenario, as shown in Fig. 2a, the Proposed framework exhibits broader coverage of the objective space, with solutions spanning a more diverse set of energy–cost trade-offs. The visual inspection of the Pareto fronts confirms the quantitative findings reported in Table 1, with the Proposed strategy achieving superior hypervolume and better exploration of high-AEP regions. The Baseline strategy, while achieving lower minimum cabling costs, shows more limited coverage of the Pareto-optimal region, particularly in high-energy production regimes. [AWAITING DATA:"Pareto front analysis","36 and 64 turbines"]

These results confirm that hierarchical problem decomposition enhances the algorithm's ability to resolve conflicting aerodynamic and electrical objectives, a property that becomes increasingly important as the number of decision variables grows.

4.3 Convergence Behavior and Search Dynamics

Figure 3 illustrates the hypervolume evolution for the Baseline and Proposed strategies at different problem scales, serving as a proxy for convergence and solution diversity.

For the 16-turbine case, as shown in Fig. 3a, the Baseline NSGA-II shows a moderate convergence rate, with the mean hypervolume increasing from approximately 7.7 to 8.5×10^{12} over 2000 generations.



(a) 16 turbines

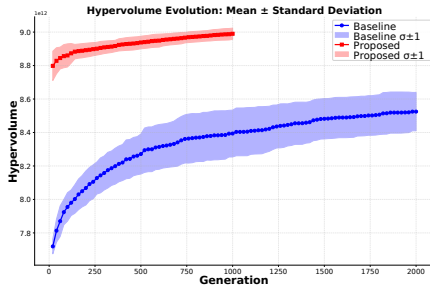
img/pareto_compacto_36.pdf

(b) 36 turbines

img/pareto_compacto_64.pdf

(c) 64 turbines

Figure 2: Pareto front comparison between Baseline NSGA-II, Sequential optimization, and the Proposed framework for increasing problem scales. The Sequential strategy converges to a single solution per run (multiple points in the aggregated plot represent different runs, not a Pareto front). For 16 turbines, the Proposed framework achieves superior hypervolume and broader coverage of high-AEP regions, indicating improved exploration of the energy–cost trade-off space. [AWAITING DATA:"Caption details","36 and 64 turbines"]



(a) 16 Turbinas

img/hv_36.pdf

(b) 36 Turbinas

img/hv_64.pdf

(c) 64 Turbinas

Figure 3: Hypervolume evolution across different farm scales. As the number of turbines increases (16 → 64), the performance gap between the Proposed strategy (red) and the Baseline (blue) widens, demonstrating the scalability of the hierarchical approach.

The wider standard deviation band, particularly in early to mid-generations (around 250–1000), indicates higher variability across independent runs, reflecting the challenges of simultaneously optimizing layout and electrical design variables in the coupled search space. Similar convergence challenges in high-dimensional multi-objective optimization have been observed in prior evolutionary studies [10, 20].

In contrast, the Proposed two-phase framework exhibits superior and more consistent convergence behavior. The Proposed strategy starts at a significantly higher initial hypervolume (approximately 8.7×10^{12}) and maintains a consistent upward trend, reaching close

to 9.0×10^{12} by generation 1000. The narrower standard deviation band, especially in later generations, indicates more robust and consistent performance across runs. This behavior reflects the staged approach: Phase 1 rapidly identifies energetically favorable layouts, while Phase 2 systematically refines cabling topology and infrastructure cost with limited Net AEP degradation. This staged convergence behavior aligns with principles from hierarchical and staged optimization literature [15, 24].

As the number of turbines increases to 36 and 64, [AWAITING DATA: "scalability analysis of convergence behavior"; 36 and 64 turbines"]. The Baseline strategy is expected to show slower convergence and reduced improvement rates, with even wider variability bands, indicating increasing difficulty in navigating the enlarged coupled search space. In contrast, the Proposed framework is expected to maintain consistent convergence behavior with narrow variability, demonstrating the scalability benefits of hierarchical decomposition.

4.4 Computational Efficiency

Computational efficiency is a critical factor for practical deployment of optimization algorithms, especially as problem scales increase. Figure 4 compares the average execution time per run across the three strategies for increasing problem scales.

For the 16-turbine scenario, the Proposed framework achieves a 31.6% reduction in execution time compared to Baseline (414 vs 606 seconds per run, averaged over 20 independent runs). This efficiency gain reflects the benefits of hierarchical decomposition: the first phase focuses on layout optimization in a reduced search space, while the second phase benefits from high-quality initial solutions provided by Smart Seeding. The Sequential strategy is the fastest (311 seconds), but this efficiency comes at the cost of solution quality and trade-off exploration, as it converges to a single solution per run. [AWAITING DATA: "Computational efficiency analysis"; 36 and 64 turbines"]

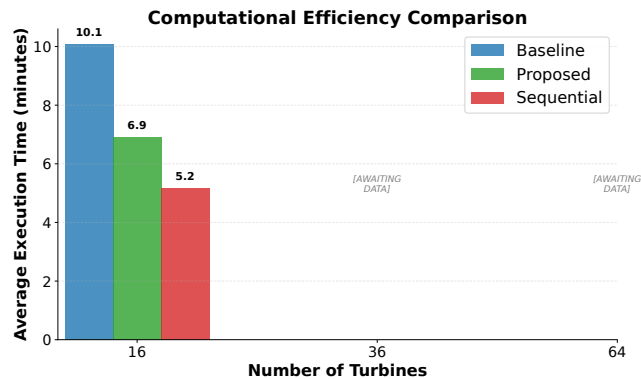


Figure 4: Average execution time per run across optimization strategies for increasing problem scales. The Proposed framework achieves significant efficiency gains compared to Baseline, while maintaining superior solution quality. [AWAITING DATA: "Execution time figure"; 36 and 64 turbines"]

As problem dimensionality increases, computational efficiency becomes increasingly important for practical applications. The hierarchical decomposition strategy demonstrates scalability advantages that are expected to become more pronounced at larger scales.

4.5 Qualitative Analysis of Solution Characteristics

Analysis of the aggregated results from 20 independent runs reveals distinct qualitative patterns in how each optimization strategy approaches the co-design problem. These patterns provide insights into the underlying mechanisms driving the observed performance differences, as summarized in Table 2.

For the 16-turbine scenario, the Baseline strategy positions the substation relatively close to the farm center (mean distance: 325 m), resulting in shorter cable lengths (mean: 13.1 km) and lower electrical losses (0.30% of gross AEP). This strategy favors cost minimization through infrastructure compactness, using 6–8 cable groups with uniform cable sections (70 or 95 mm²). However, this compact configuration constrains layout flexibility, limiting exploration of high-AEP regions. The analysis reveals that this cost-focused approach trades off spatial freedom for infrastructure efficiency, resulting in solutions that cluster in lower-AEP regions of the objective space.

In contrast, the Proposed framework exhibits a fundamentally different approach. The substation is positioned significantly farther from the center (mean distance: 2080 m), enabling better exploitation of the available site area for turbine placement optimization. This strategic positioning results in longer cable lengths (mean: 39.0 km) and higher electrical losses (0.93% of gross AEP), but these infrastructure costs are offset by superior energy production. The Proposed strategy uses fewer cable groups on average (mean: 6, range: 4–9) and explores a wider variety of cable sections (70, 95, and 120 mm²), reflecting more sophisticated trade-off exploration between cable sizing and routing efficiency. This finding suggests that hierarchical decomposition enables the algorithm to discover non-obvious design configurations where infrastructure investments are strategically allocated to maximize overall system performance.

The Sequential strategy, while achieving the highest AEP (463.0 GWh), incurs the highest infrastructure costs (mean: 731k USD) due to its fixed-layout constraint. With the turbine positions frozen from Phase 1, the subsequent electrical optimization can only adjust substation location and cable grouping, resulting in suboptimal infrastructure configurations that fail to exploit layout–infrastructure coupling. This observation reveals a critical limitation: when layout decisions are made independently of infrastructure considerations, the resulting solutions exhibit structural inefficiencies that cannot be corrected in a subsequent optimization phase.

These qualitative differences highlight how hierarchical decomposition enables the Proposed framework to explore a broader design space, making strategic trade-offs between energetic and infrastructure objectives that are inaccessible to both the fully coupled Baseline approach and the decoupled Sequential strategy. [AWAITING DATA: "Qualitative analysis"; 36 and 64 turbines"]

As problem scale increases to 36 and 64 turbines, [AWAITING DATA: "scalability analysis of qualitative characteristics"; 36 and 64 turbines"]. The relative importance of substation positioning, cable grouping strategies, and electrical loss management is expected to evolve, with infrastructure decisions becoming increasingly critical as farm density increases. The Proposed framework's ability to co-optimize these variables is expected to provide even greater

advantages at larger scales, where the coupling between layout and infrastructure becomes more pronounced.

4.6 Representative Layout Examples

Figure 5 presents representative knee-point solutions obtained from a single representative run (one of 20 independent runs) for illustrative purposes. The layouts exemplify the distinct design patterns emerging from each optimization strategy, visually confirming the qualitative differences discussed in the previous section.

For the 16-turbine scenario, the Baseline strategy favors compact infrastructure near the center, constraining spatial exploration. The Proposed framework positions the substation farther from the center, enabling better exploitation of the available site area and superior energy production despite longer cable lengths. The Sequential strategy achieves the highest AEP but incurs the highest infrastructure costs due to fixed-layout constraints that cannot be corrected in Phase 2. These visual examples demonstrate how hierarchical decomposition enables discovery of non-obvious design configurations where strategic infrastructure positioning maximizes overall system performance. [AWAITING DATA: "Layout examples analysis"; 36 and 64 turbines"]

4.7 Discussion and Limitations

The comparative analysis presented in this study is based on standardized scenarios derived from the IEA Task 37 benchmark [1], enabling controlled and reproducible comparisons across optimization strategies. This abstraction omits certain site-specific constraints (e.g., bathymetry variations, exclusion zones, port accessibility). The uniform cable section strategy adopted here reflects standard industrial practice, where a single cable cross-section is typically selected for the entire collection system to simplify procurement and installation.

The results demonstrate that hierarchical evolutionary strategies significantly enhance robustness, scalability, and interpretability in offshore wind farm co-design across the tested problem scales. The comparison is restricted to NSGA-II as the underlying multi-objective algorithm, and alternative evolutionary paradigms (e.g., MOEA/D, SPEA2) may exhibit different scalability characteristics when integrated with the hierarchical framework.

Future work will extend the framework to heterogeneous cable cross-sections, larger turbine counts, and site-dependent constraints, as well as explore adaptive phase-transition mechanisms to further improve scalability in industrial-scale applications. Investigation of alternative multi-objective evolutionary algorithms and their integration with the hierarchical framework will also be pursued.

5 Conclusion

This paper investigated how different evolutionary optimization strategies influence the quality, scalability, and robustness of integrated offshore wind farm design. Rather than proposing a new evolutionary algorithm, the study focused on a systematic comparison between three paradigms: a fully coupled single-phase NSGA-II, a sequential layout-infrastructure optimization strategy, and a hierarchical two-phase evolutionary framework.

The results demonstrate that the optimization strategy itself plays a decisive role in the ability to resolve trade-offs between aerodynamic performance and electrical infrastructure cost. While the Sequential approach converges rapidly, it consistently collapses to a single solution due to its fixed-layout assumption, preventing meaningful exploration of the Pareto front. The Baseline NSGA-II, although capable of generating non-dominated solutions at small scales, exhibits degraded convergence and reduced Pareto coverage as problem dimensionality increases.

In contrast, the proposed two-phase framework achieves stable and well-distributed Pareto fronts across all tested scales (16, 36, and 64 turbines). By decoupling energetic layout exploration from electrical infrastructure refinement, the framework maintains robustness to stochastic effects, preserves solution diversity, and scales more effectively with increasing numbers of decision variables. The deterministic angular partitioning strategy further ensures feasible, planar cable topologies without geometric repair, enabling efficient integration of electrical constraints within the evolutionary loop.

The main contributions of this work are threefold: (i) a two-phase hierarchical optimization strategy that leverages mono-objective layout exploration to guide subsequent multi-objective co-optimization, (ii) a phase-bridging population initialization mechanism (Smart Seeding) that seeds high-quality layouts while diversifying substation locations to preserve exploration, and (iii) a deterministic angular sector grouping strategy that enforces planar, non-overlapping cable strings without the need for geometric repair heuristics.

From an application-oriented perspective, the framework provides decision-makers with explicit trade-offs between Net AEP and cabling CAPEX, rather than single-point solutions. This capability is particularly relevant for early-stage offshore wind farm planning, where uncertainty is high and flexibility in design choices is essential. The results suggest that hierarchical evolutionary strategies offer a practical pathway for bridging aerodynamic and electrical considerations in large-scale offshore wind farm co-design.

References

- [1] Nicholas F. Baker, Andrew P. Stanley, Jared J. Thomas, Andrew Ning, and Katherine Dykes. 2019. Best practices for wake model and optimization algorithm selection in wind farm layout optimization. In *AIAA Scitech 2019 Forum*. 0540. doi:10.2514/6.2019-0540 Defines the IEA Task 37 benchmark cases..
- [2] Majid Bastankhah and Fernando Porté-Agel. 2014. A new analytical model for wind-turbine wakes. *Renewable Energy* 70 (2014), 116–123. doi:10.1016/j.renene.2014.01.002
- [3] M. Bastankhah and F. Porté-Agel. 2016. Experimental and theoretical study of wind turbine wakes in yawed conditions. *Journal of Fluid Mechanics* 806 (2016), 506–541. doi:10.1017/jfm.2016.595
- [4] Italo Firmino da Silva, Telles Brunelli Lazzarin, Lenon Schmitz, and Alison R. Panisson. 2025. Genetic Algorithm-Based Optimization Framework for Offshore Wind Farm Layout Design. *IEEE Access* 13 (2025), 170081–170094. doi:10.1109/ACCESS.2025.3614516
- [5] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan. 2002. A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE Transactions on Evolutionary Computation* 6, 2 (2002), 182–197.
- [6] M. Fischetti, J. Leth, and A. B. Borchersen. 2015. A Mixed-Integer Linear Programming approach to wind farm layout and inter-array cable routing. In *American Control Conference*. 7172266. doi:10.1109/ACC.2015.7172266
- [7] Randy L. Haupt and Sue Ellen Haupt. 2004. *Practical Genetic Algorithms* (2nd ed.). John Wiley & Sons. Essential for justifying continuous-parameter encoding in hybrid optimization..
- [8] IRENA. 2024. Renewable Power Generation Costs in 2023. International Renewable Energy Agency, Abu Dhabi.
- [9] N. O. Jensen. 1983. *A note on wind generator interaction*. Technical Report Risø-M-2411. Risø National Laboratory, Roskilde, Denmark.

Table 2: Qualitative characteristics of optimization strategies across problem scales (mean $\pm$ std over 20 independent runs).

Turb.	Characteristic	Baseline	Proposed	Sequential
16	Substation distance [m]	325 ± 153	2080 ± 971	552 ± 348
	Cable length [km]	13.1 ± 2.4	39.0 ± 5.5	41.9 ± 2.4
	Number of cable groups	7.0 ± 1.0	6.0 ± 1.2	6.0 ± 1.0
	Electrical losses [%]	0.30	0.93	0.90
	Cable sections [mm ²]	70, 95	70, 95, 120	70, 95, 120
36	Substation distance [m]	[AWAITING DATA]	[AWAITING DATA]	[AWAITING DATA]
	Cable length [km]	[AWAITING DATA]	[AWAITING DATA]	[AWAITING DATA]
	Number of cable groups	[AWAITING DATA]	[AWAITING DATA]	[AWAITING DATA]
	Electrical losses [%]	[AWAITING DATA]	[AWAITING DATA]	[AWAITING DATA]
	Cable sections [mm ²]	[AWAITING DATA]	[AWAITING DATA]	[AWAITING DATA]
64	Substation distance [m]	[AWAITING DATA]	[AWAITING DATA]	[AWAITING DATA]
	Cable length [km]	[AWAITING DATA]	[AWAITING DATA]	[AWAITING DATA]
	Number of cable groups	[AWAITING DATA]	[AWAITING DATA]	[AWAITING DATA]
	Electrical losses [%]	[AWAITING DATA]	[AWAITING DATA]	[AWAITING DATA]
	Cable sections [mm ²]	[AWAITING DATA]	[AWAITING DATA]	[AWAITING DATA]

- [10] Z. Jiang, H. Li, et al. 2025. Review of Artificial Intelligence-Based Design Optimization of Wind Power Systems. *Wind* 5, 3 (2025).
- [11] L. Jiangyi, H. Huo, and Z. Qian. 2025. Topology Optimization of Offshore Wind Farm Collection System Based on Priority Queue Esau-Williams Algorithm. In *Lecture Notes in Electrical Engineering*, Vol. 1397. doi:10.1007/978-981-96-4059-1_26
- [12] R. Jin, P. Hou, Y. Qi, et al. 2025. Joint planning of distribution and transmission system for offshore wind farms integrated with hydrogen production platforms. *Applied Energy* 398 (2025). doi:10.1016/j.apenergy.2025.126383
- [13] R. Jin, P. Hou, G. Yang, Y. Qi, C. Chen, and Z. Chen. 2019. Cable routing optimization for offshore wind power plants via wind scenarios considering power loss cost model. *Applied Energy* 254 (2019), 113719. doi:10.1016/j.apenergy.2019.113719
- [14] Ivan Katic, Jørgen Højstrup, and Niels Otto Jensen. 1986. A simple model for cluster efficiency. In *European Wind Energy Association Conference and Exhibition*. A.D. Stephens, 407–410.
- [15] Steven M. LaValle. 2006. *Planning Algorithms*. Cambridge University Press. Provides the formal definition of Configuration Spaces for multi-body systems..
- [16] M. O. MacHado, I. F. D. C. Fernandes, et al. 2024. A hybrid multi-agent meta-heuristic for the offshore wind farm cable routing problem. *Expert Systems with Applications* 255 (2024). doi:10.1016/j.eswa.2024.124668
- [17] W.-S. Moon, J.-N. Won, J.-S. Huh, A. Jo, and J.-C. Kim. 2015. Optimal design of power grid and location of offshore substation for offshore wind power plant. *Journal of the Korean Institute of Electrical Engineers* 64, 7 (2015), 984. doi:10.5370/KIEE.2015.64.7.984
- [18] Moussa Nakhai et al. 2023. Electrical Collection System Cost Modeling for Marine Renewable Energy Farms. *Journal of Marine Science and Engineering* 11, 7 (2023), 1348.
- [19] Jorge Nocedal and Stephen J. Wright. 2006. *Numerical Optimization* (2nd ed.). Springer. Standard reference for defining optimization variables within R^n manifolds..
- [20] T. A. Qureshi and V. Warudkar. 2023. Wind farm layout optimization through optimal wind turbine placement using a hybrid PSO-GA. *Env. Science and Pollution Research* 30 (2023).
- [21] X. Shen, Q. Wu, H. Zhang, and L. Wang. 2023. Optimal Planning for Electrical Collector System of Offshore Wind Farm With Double-Sided Ring Topology. *IEEE Transactions on Sustainable Energy* 14 (2023). doi:10.1109/TSTE.2023.3241357
- [22] Italo Firmino Silva, Telles B. Lazzarin, Lenon Schmitz, and Alison R. Panisson. 2025. Optimizing Wind Farm Project Assessments Using Genetic Algorithms. In *Proceedings of the Genetic and Evolutionary Computation Conference Companion* (Malaga, Spain) (GECCO '25 Companion). Association for Computing Machinery, New York, NY, USA, 811–814. doi:10.1145/3712255.3726546
- [23] Mauricio Souza de Alencar, Tuhfe Göcmen, and Nicolaos A. Cutululis. 2026. Flexible cable routing framework for wind farm collection system optimization. *European Journal of Operational Research* 329, 10 (2026), 1037–1051. doi:10.1016/j.ejor.2025.07.069
- [24] S. Tao, J. Yang, et al. 2025. A two-stage optimization model for floating offshore wind farm layout and control. *Renewable Energy* 253 (2025).
- [25] L. Wang, J. Wu, et al. 2019. An integration optimization method for power collection systems of offshore wind farms. *Energies* 12 (2019).
- [26] J. Ye, G. Zhou, et al. 2023. Path Optimization of Submarine Cables for Offshore Wind Farm Considering Feeder Crossing Avoidance. *Automation of Electric Power Systems* 56 (2023).
- [27] S. Yuan and K. Tang. 2025. Application of an Improved Ant Colony Optimization Algorithm to Wind-Farm Topology Design. In *IEEE International Conference on Signal Processing*. doi:10.1109/ICSP65755.2025.11086778

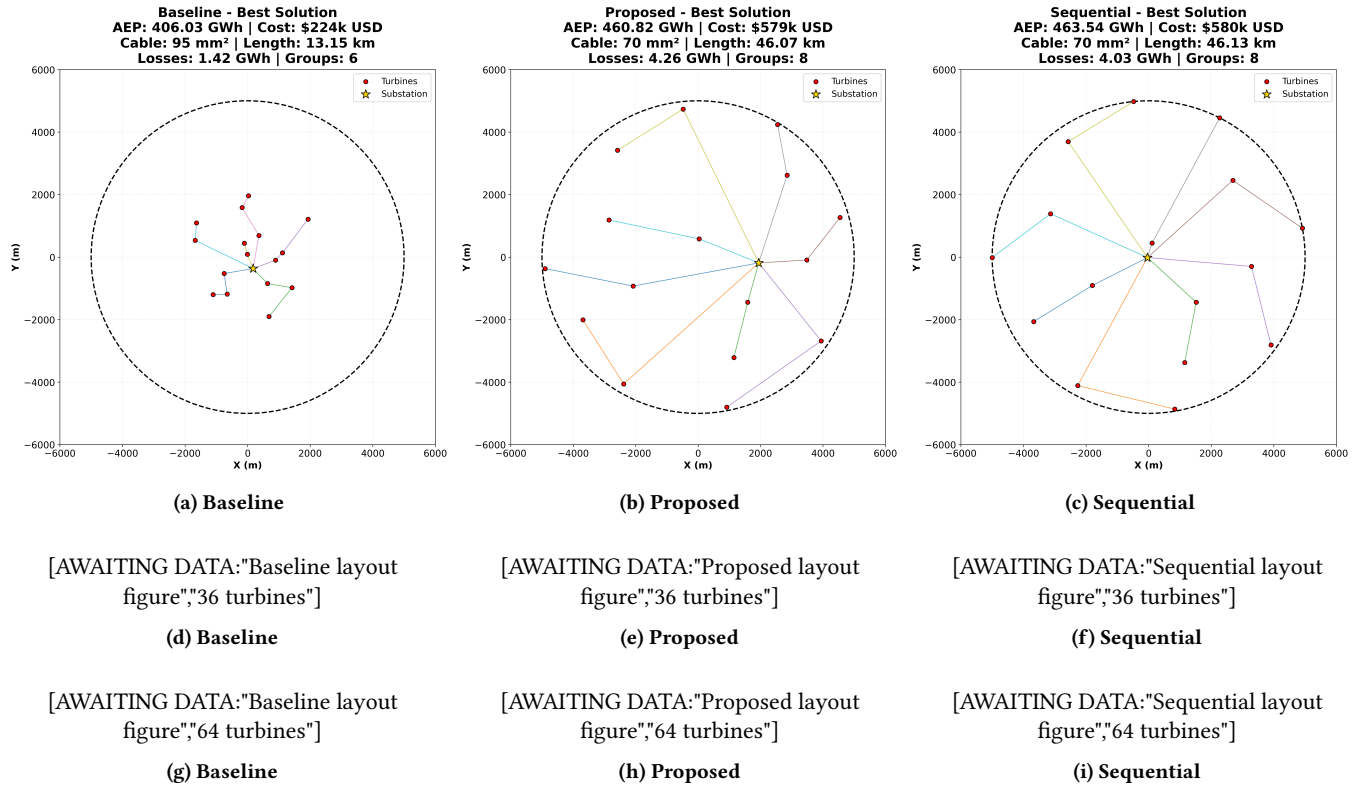


Figure 5: Representative knee-point solutions across problem scales, obtained from a single representative run (one of 20 independent runs) for illustrative purposes. Top row: 16 turbines; middle row: 36 turbines; bottom row: 64 turbines. Columns represent Baseline, Proposed, and Sequential strategies, respectively. Colors denote cable strings generated by angular partitioning.